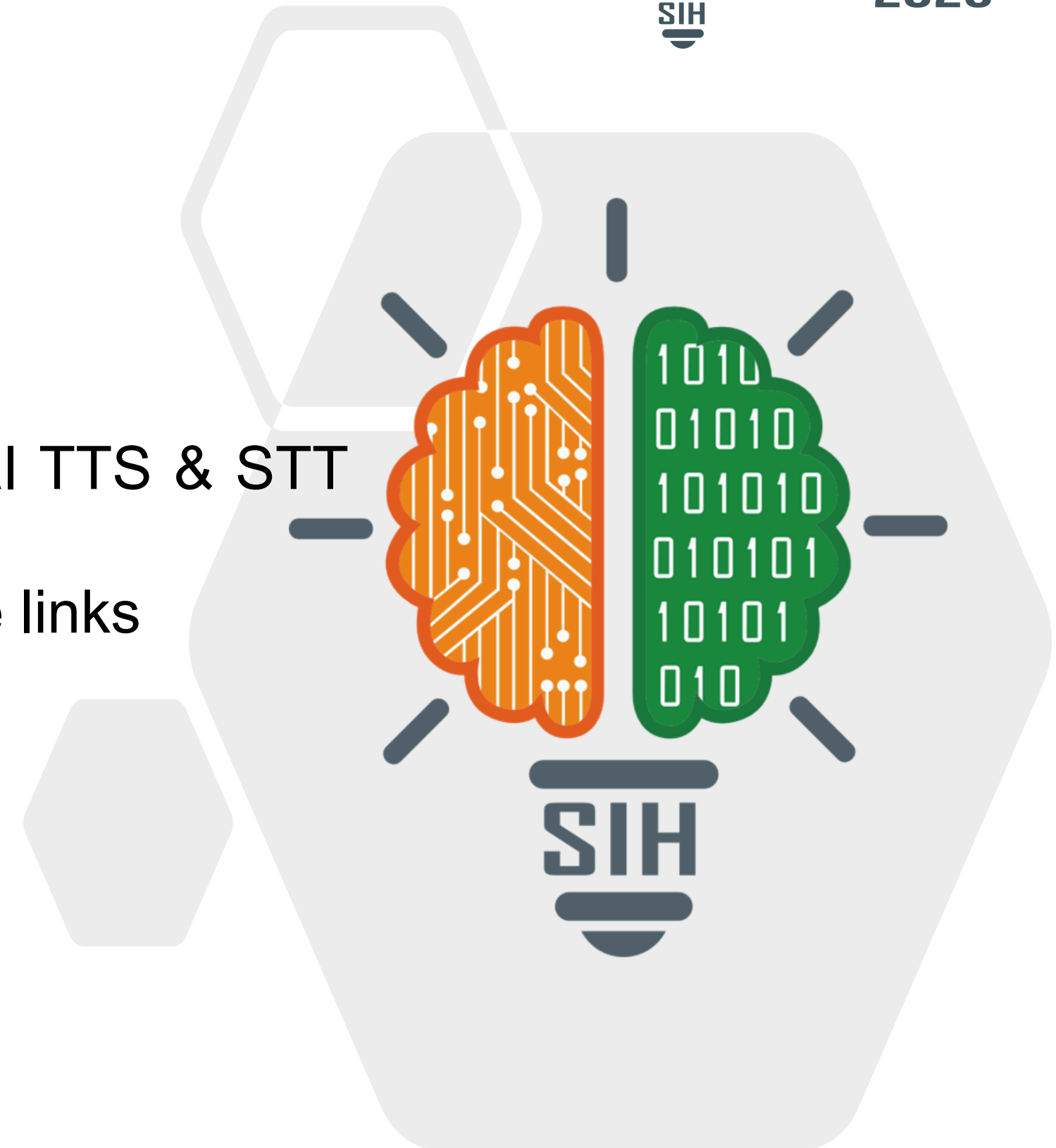


TITLE PAGE

- **Problem Statement ID –26173**
- **Problem Statement Title-** iTantra -Indian Multilingual TTS & STT

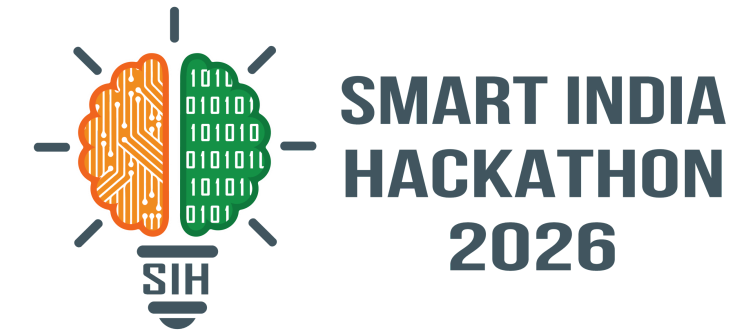
Aided Neural Transceiver Radio Access for low bitrate links

- **Theme-**Smart Automation
- **PS Category-** Software
- **Team ID-**143156
- **Team Name-** StepX



StepX

SANWAD



OUR SOLUTION



- A push-to-talk app for two phones with no internet connection between them.
- One phone listens to speech and turns it into text, sends that text over Wi-Fi or Bluetooth, the other phone reads it aloud.
- It supports 10 Indian languages, so the message stays in the speaker's own language from one end to the other.
- Emergency alerts play at full volume and cannot be silenced.

HOW IT IS DIFFERENT

- Bhashini, Google, Zello — need the internet
- Bridgefy, Briar, Meshtastic — work offline, but only send text you must read
- FreeDV, Codec2 radios — send voice, but stop at 700 bps and speak no Indian language
- Ours is voice in and voice out, offline, in 10 Indian languages.

WHY IT IS DIFFERENT

- Sending your actual voice needs roughly 8,000 to 16,000 bits every second.
- Emergency and satellite links often carry only a few hundred bits per seconds, raw Voice simply doesn't fit.
- One spoken sentence written down is about 118 bytes.
- We squeeze it to 44 and wrap it up to about 100 bytes. That fits almost any link.

HOW IT WORK

1 Hold the button and Speak

Push-to-talk, in any of 10 Indian languages.



2 The phone writes it down

Speech becomes text the moment you release.

3 The words travel



Bluetooth, Wi-Fi Direct, or a cable into a radio.

4 Other phone

Checked, repeats dropped, sentence rebuilt.



5 It speaks out loud

A natural voice, in the language you used.

6 If it is an emergency

Full volume, ignores silent mode, cannot be stopped.

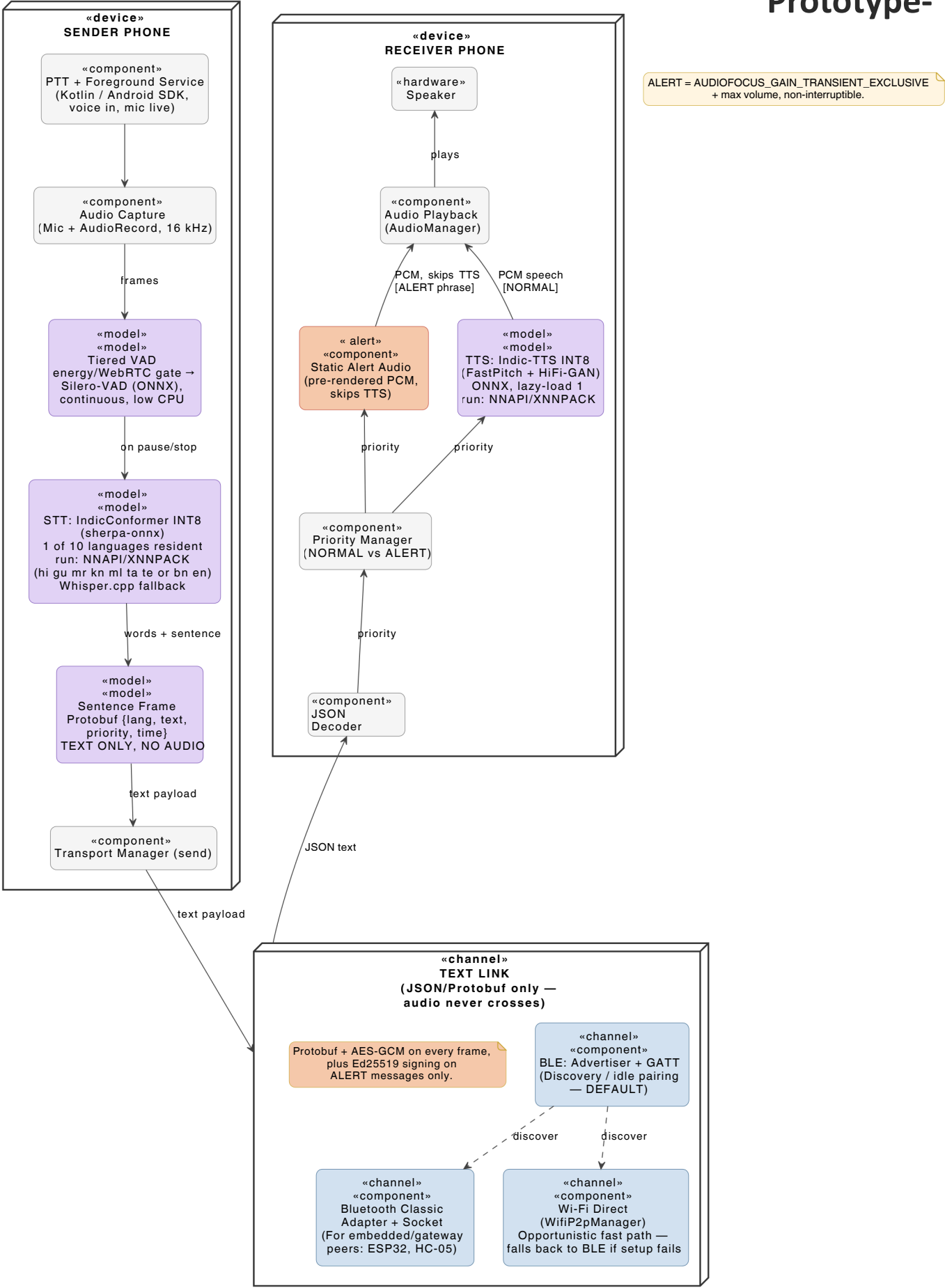
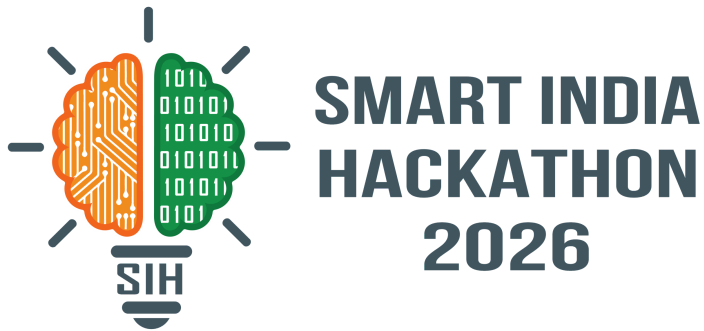


Just speak and listen — nobody ever has to read.

StepX

TECHNICAL APPROACH

Prototype- <https://stepx-work.github.io/sanwad/>



Tecnologies to be Used

APP

Kotlin • Jetpack Compose • MVVM • Coroutines

LOCALIZATION

UI strings per locale (10 languages) • native-script language picker

SPEECH IN (VAD + STT)

Tiered VAD: energy/WebRTC gate → Silero-VAD • IndicConformer INT8 (sherpa-onnx) • Whisper.cpp fallback

SPEECH OUT (TTS)

Indic-TTS INT8 (FastPitch + HiFi-GAN) • pre-rendered alert bank • espeak-ng fallback

ML RUNTIME

Wi-Fi Direct / local Wi-Fi • Bluetooth (BLE + RFCOMM)

COMMS

BLE (default) • RFCOMM (embedded/gateway peers) • Wi-Fi Direct (opportunistic fast path)

TRANSPORT

UDP for chat • TCP for guaranteed alerts

DATA SECURITY

Protobuf frames + AES-GCM • Ed25519 signing (alert path only) • Room/SQLite

AUDIO TUNING

AudioRecord/AudioTrack • INT8/FP16 • AEC (full-duplex) • latency logger

TECHNICAL

- Every part open-source & verified.
- INT-8 quantized models (42–188 MB/lang) fit within 2 GB RAM budget.
- Utterance-level non-streaming decode prevents hallucination & lowers CPU.

ECONOMIC

- ₹0 infrastructure — no towers, servers, or cellular data.
- Low standby footprint: Silero-VAD/TFLite gate draws <15% idle CPU.
- Targets ₹8–10k · 3-4 GB RAM phones.

OPERATIONAL

- Voice-first UX for illiterate users; 1-tap zero-pairing walkie-talkieders.
- End-to-end voice relay in seconds over peer-to-peer WIFI/Bluetooth
- Non-interruptible alerts bypass system.

Potential challenges and risks

IndicConformer has no ready Odia export



Convert/distill from the AI4Bharat checkpoint before demo day; disclose the gap openly if not ready.

OEM ROMs kill mic & alarm playback



Foreground service + alarm stream + whitelist + ROM test script.

Phone radios reach only ~100 m



Mesh relay + LoRa bridge + SMS + store-and-forward.

Noisy venues · ta/ml/te WER fairness



Noise-robust VAD + published char-level WER protocol.

36-hour offline finale build



Vendored deps; pre-paired spare phones; rehearsed demo kit.

Bundling all 10 languages = ~1.5 GB bloat



On-demand packs — install only your preferred languages.

Strategy pillar	Application to sanwad
Target End-Users	NDRF (Disaster Response), State Maritime Boards (Fishermen), and rural frontline workers operating beyond cell coverage.
Distribution Channel	B2G (Business-to-Government) handoff. The app serves as a Digital Public Infrastructure (DPI) tool distributed via government portals or pre-installed on state-subsidized hardware.
Financial viability	Hyper-sustainable because of the ₹0 recurring operational cost. The government pays zero cloud or cellular transmission fees since all computing is done on the edge.
Maintenance Model	Open-source community stewardship, with language-model updates tracked against upstream AI4Bharat / sherpa-onnx releases.

Potential impact



- **Fishermen & coastal communities** — SOS without towers, in their own language
- **Disaster relief** — village-wide alerts when networks are down
- **Elderly & low-literacy users** — nothing to read or type, ever
- **Disabled users** — live transcript for deaf users; voice output for users who can't easily type.
- **Remote teams** — border posts, expeditions, ISRO low-bitrate operations

Benifits

Strategy pillar	Application to sanwad
SOCIAL	-Language Inclusivity: Full coverage for 9 Indic languages + Indian English, breaking linguistic barriers in diverse rescue zones. Life-Safety Alert Policy: Emergency frames bypass mute/DND via USAGE_ALARM at maximum volume to wake operators even from locked screens.
ECONOMIC	₹0 Infrastructure & Transmission: Zero cellular airtime, zero cloud compute, and zero recurring server costs. Runs on Budget Hardware: Fully optimized via int8 quantization for ultra-budget ₹8,000–₹10,000 Android phones (Cortex-A53/A55, 4 GB RAM).
ENVIRONMENT · STRATEGIC	Runs on Budget Hardware: Fully optimized via int8 quantization for ultra-budget ₹8,000–₹10,000 Android phones (Cortex-A53/A55, 2 GB RAM). Low Battery Footprint: Silero-VAD gate draws <15% single-core CPU during idle listening, giving 8+ hours of armed standby.

50–100× lighter than audio codecs	≤ 2–3 s mouth-to-ear latency	10 Indian languages, offline	₹0 network / infra cost	8 h+ armed standby battery
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Capability / Feature	SANWAD (Proposed)	Standard Walkie-Talkie (VHF)	Opus over WebRTC
Required Channel Bitrate	~0.5 – 1.0 kbps (Text)	Analog (Broadband RF)	12 – 32 kbps (Compressed)
Network Infrastructure	Zero (Ad-Hoc P2P)	Direct Line-of-Sight	Local Wi-Fi access point required
Cross-language speech	Yes (Local STT & TTS)	No (Raw Audio Only)	No (Raw Audio Only)
Non-Interruptible Max Alert	Yes (USAGE_ALARM Override)	Manual Volume Knob	Subject to Media Volume
Device Power Consumption	Ultra-Low (VAD-gated BLE scan)	High Continuous RF Drain	Moderate

- Whisper, Radford et al., 2022 — <https://arxiv.org/abs/2212.04356>
- Conformer, Gulati et al., Interspeech 2020 — arxiv.org/abs/2005.08100
- VITS, Kim et al., ICML 2021 — arxiv.org/abs/2106.06103
- IndicConformer (MIT) — github.com/AI4Bharat/IndicConformerASR
- IndicConformer 120M CTC ONNX — huggingface.co/trysem/indicconformer-120m-onnx
- IndicF5 — huggingface.co/ai4bharat/IndicF5
- Silero-VAD — github.com/snakers4/silero-vad
- sherpa-onnx — github.com/k2-fsa/sherpa-onnx
- ONNX Runtime Mobile — onnxruntime.ai/docs/tutorials/mobile
- RFC 4838, Delay-Tolerant Networking — rfc-editor.org/rfc/rfc4838
- ggwave (data-over-sound) — github.com/ggerganov/ggwave